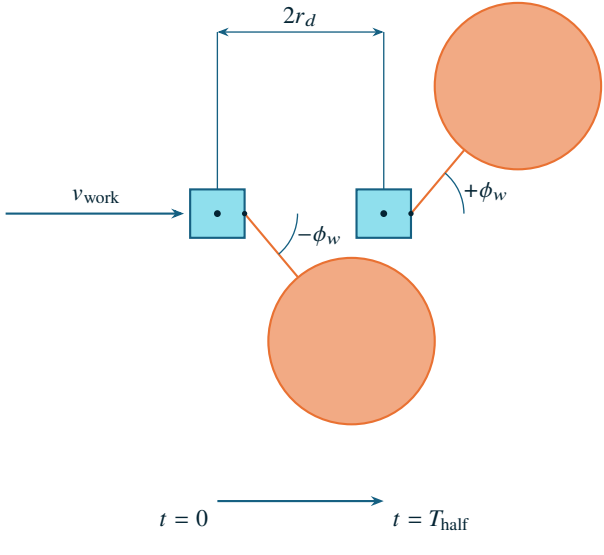

$2r_d$
v_{work}
$-\phi_w$
$+\phi_w$
$t = 0$
$t = T_{\text{half}}$